

What are the various types of AI?	
Explain the PEAS description for Automated taxi driver.	
Write a short note on Goal based agent with a diagram.	
Differentiate between supervised and unsupervised Learning.	
Give the difference between informed search and uninformed search algorithms.	
Explain problem formulation steps with water jug problem in which a full 4-gallon jug and an empty 3-gallon jug, the goal is to fill the 4 gallon jug with exactly 2- gallons of water.	
What is an adversarial search algorithm and a game playing algorithm?	
List out foundation of AI and describe any two in detail.	
Illustrate application areas of AI.	
What is a PEAS description. Give peas description for a robot meant for cleaning the house?	
Define pruning? Explain alpha and beta pruning algorithm with an example.	
Explain all the AI agent with diagram.	
Write a short note on NLP.	
What are Problems Robots can solve.	
Describe levels of knowledge based agents.	
Write a short note on Breadth First Search.	
Define uncertainty in AI. What are the sources of uncertainty in real-world problems?	
Explain the components of the AI Program.	
Describe different forms of learning.	
Illustrate application areas of AI.	
Classify the types of chaining.	
Explain various properties of the task environment with suitable examples.	
Elaborate Heuristic functions of 8 tile puzzle problem.	
To demonstrate minimax with alpha-beta pruning, we use the following data to generate tree. (((3 17) (2 12)) ((15) (25 0))) (((2 5) (3)) ((2 14))))	
Describe in detail a problem solving agent with a diagram. What is the difference between Problem solving agent and Planning agent?	
Illustrate forward chaining and backward chaining algorithms with suitable examples.	
Define Belief Network. Describe the steps of constructing a belief network with an example. What types of inferences can be drawn from that?	
Explain the different types of environments of AI agents.	
Explain Hill climbing algorithm and problems that occur in hill climbing algorithm?	
Explain Genetic Algorithms in details. explain it with an example.	

Distinguish between Propositional logic (PL) and first order predicate logic (FOPL) knowledge representation mechanisms. Take suitable examples for each point of differentiation.

What are different types of Planning? Explain any two types with an example?

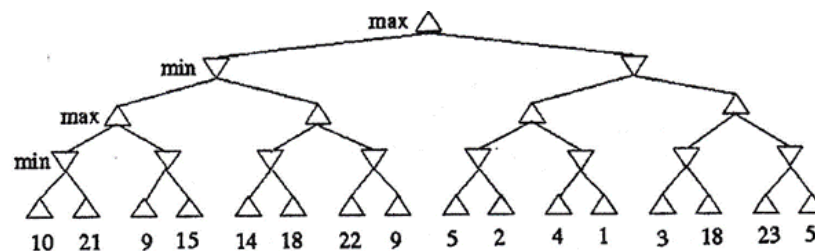
Discuss A* Algorithm with an example.

Describe PEAS description. Illustrate the PEAS description for Vacuum Cleaner Agent.

What is reinforcement learning? Explain its types.

Describe Bayesian Belief Network in detail.

Apply Alpha-Beta Pruning on the following example given in Figure considering the first node as max.



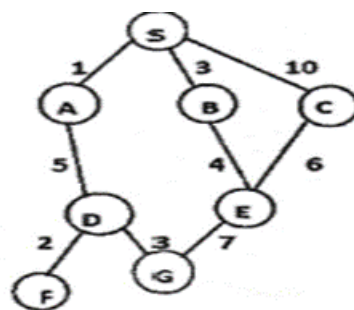
Explain Partial order planning with suitable examples.

Explain different applications of AI in healthcare, retail and banking.

Differentiate between first-order logic and propositional logic.

Write short note on PAC Learning

Consider the graph given in Figure below. Assume that the initial state is S and the goal state is G. Show how A* Search would create a search tree to find a path from the initial state to the goal state:



Write a short note on Statistical learning.

Explain the concept of hill climbing algorithm & simulated annealing algorithm.